SALIENCY MAPS AND INPUT-SPACE INTERPRETABILITY

STUDY NOTES AND QUICK REFERENCE

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INTRODUCTION

Deep neural networks, especially those that exist in computer vision, have achieved very strong performances in multiple tasks including image classification, object detection and medical diagnosis. However, despite their accuracy these models are often difficult to interpret due to their complex and highly non-linear structure. This raises concerns in domains where understanding how a model made a decision is just as important as the decision itself. To be able to understand what parts of an image influence the decision of a neural network, techniques such as **input gradients** and **saliency maps** are used. These methods act as a window into the model's black box logic, helping us see which pixels it attended to when making predictions.

Saliency-based interpretability methods provide visual explanations for the behaviour of the model. By analysing the relationship between the small changes in the input pixels and the output, these techniques can estimate the importance of each pixel and present it in the form of a saliency map. This allows us to find out which **parts of an image the model is focusing on when making predictions**.

The idea of saliency comes from how a human sees certain objects. Our eyes look at certain parts that naturally grab our attention because they look different. By studying how humans process visual data, early models—like the one from Itti et al.—were able to map out how we decide what's worth looking at. **Modern deep learning interpretability** has shifted from edge-based features to gradient-based methods, such as **vanilla saliency**, **guided backpropagation**, and **SmoothGrad**, to directly calculate feature importance. These techniques are used to analyse network decisions, revealing both the strengths and limitations of how models process information.

In this topic, input-space interpretability methods will be explored, focusing on input gradients, vanilla saliency maps, guided backpropagation, and SmoothGrad. The way these techniques work together with their strengths and limitations, will be looked at. Moreover, the way on how they can be used to better understand the decision-making process of neural networks will be tackled. .

BIOLOGICAL FOUNDATIONS

Saliency is the part of an object that stands out. This concept emerged from studies of human visual attention. Humans must visually process large amounts of information efficiently. Therefore, the human brain relies on a bottom-up mechanism that rapidly highlights visually distinctive regions for further analysis.

Itti et. al. defined saliency using a **pre-attentive, bottom-up approach**. This was designed to simulate how the primate visual system handles information bottlenecks by **rapidly selecting conspicuous locations for detailed analysis**. An image is decomposed into multiple feature channels, including intensity, colour, and orientation. The model would analyse these features at different spatial dimensions to capture both fine details and broader structures. By measuring centre-surround contrasts, the models identify exactly how a region pops against its background. This data would produce feature maps that showcase high-contrast areas. The model would then generate a final saliency map by combining these feature maps, with the highest numerical values representing the most visually dominant regions. This map indicates where the visual system should focus next, **simulating both rapid detection and sequential attention shifts**.

However, when deep neural networks were introduced, the classical saliency method could not be directly applied to such networks. This was because these methods were designed to detect low-level features such as edges and textures that human eyes perceive. To combat this, **deep learning saliency techniques** were employed. These are **post-hoc interpretability tools** (attribution methods) that are used to identify which input features (pixels) massively influenced a neural network's final classification score.

INPUT GRADIENTS

An input gradient is a measure of the **tiny changes in a specific pixel** which can **impact the network's final confidence score**. Imagine standing in a room in front of a wall that contains multiple tiny dimmer switches, where each switch represents a pixel in an image. Somewhere else, there is a meter showing how confident the computer is deducing that the image is a cat. An input gradient is when you ask the question: "If I slowly push this tiny switch, how much will that Cat need to move?"

If moving a pixel's value (its brightness or color) makes the "Cat" score jump significantly, the network would be highly sensitive towards that pixel. If moving the pixel does nothing, then the network is simply ignoring that pixel.

A **saliency map** (also known as a **sensitivity or pixel attribution map**) is the result of gathering all of the tiny "nudges" and displaying them as a single image. This map works in a similar way as a heat-sensing camera for the AI's brain. When the AI looks at pictures of a gazelle, the saliency maps would highlight important areas of the gazelle such as the horns and the ears. Through these hot spots, the AI can deduce that the image can be labelled as an image representing a gazelle. A saliency map can turn the mathematical importance of each pixel into a visual guide that humans can understand. For instance, if you see high "heat" on a gazelle's body but none on the grass behind it, you know that the model is focused on the correct object rather than random background clues.

One problem that is common with basic gradients is the fact that these **gradients are susceptible to noise** where the gradients highlight random pixels that don't seem important to a human eye. This is similar to a high-precision tool reacting to the smallest disturbances. The core pattern still remains the same, but **small and random fluctuations often create a lot of random noise in the readings**. To solve this problem, techniques such as the **SmoothGrad technique** can be used. This technique would take slightly different versions of the image by adding a bit of static or noise to each image. The gradients for all of the images are then calculated and then averaged out. This technique is similar to taking 50 blurry photos of the same scene and layering them to get one sharp, clear picture.

Saliency Maps and Input Gradients are important for a network's decision making as it can help developers debug models. If a self-driving car forgets to stop in front of a fire truck, a saliency map can check if it was focused on the vehicle or on a bright sign on the side of the road. By highlighting the important pixels, the saliency map can verify whether the model is following the correct rules instead of taking shortcuts.

There are various **mathematical formulations** that are used for input gradients:

- **Vanilla Saliency Maps**

A vanilla saliency map (or sensitivity map) is found by **finding the derivative of the class score with respect to the input image**. The mathematical formula for a vanilla saliency map is as follows:

$$M_C(x) = \partial S_C(x) / \partial(x)$$

where $S_C(x)$ is the score for a class c that is computed by the network for an input x.

This equation represents how much a tiny change in each pixel of the input x affects the final classification score.

The algorithm for a vanilla saliency map works as follows

1. A forward pass through the network is performed with the input x to be able to calculate the activation function S_C for the target class c.
2. A backpropagation pass is performed to be able to find the gradient of S_C with respect to the input pixels.
3. Optionally, you can take the absolute value of the gradients or put a limit on any outlying values to create a more visually coherent heatmap.

- **Guided Backpropagation**

Guided Backpropagation is designed as an improvement to **prioritise positive contributions to the final prediction** by modifying the standard backpropagation algorithm. This technique adjusts the calculation of the input gradients at the Rectified Linear Unit (ReLU) layers. During the backward pass, it is ensured that only positive gradients that relate to positive activations are propagated.

The algorithm works as follows:

1. A standard forward pass is performed to be able to compute activations.
2. During backpropagation, at each ReLU layer, any negative gradient values are discarded.

3. Effectively, the zero-masks gradients where the gradient is negative or where the input activation was negative, focusing the visualization on features that positively triggered high-level units.

- **SmoothGrad**

SmoothGrad is a technique that is used to “sharpen” any noisy gradient-based maps by averaging the gradients of many versions of the same image perturbed with noise.

The mathematical formula for SmoothGrad is:

$$\widehat{\mathbf{M}}_c(\mathbf{x}) = 1/n \sum_{i=1}^{n} \mathbf{M}_c(\mathbf{x} + \mathbf{N}(\mathbf{0}, \boldsymbol{\sigma}^2))$$

where

n represents the number of noise trials,

$\mathbf{N(0, \sigma^2)}$ represents the Gaussian noise with standard deviation σ

$\widehat{\mathbf{M}}_c$ is the smoothed gradient of $M_c(x)$. This is defined as the average of the model’s sensitivity that is measured according to slightly distorted maps of n noisy samples.

The algorithm is as follows:

1. A sample size n (typically n=50 provides diminishing returns for higher values) and a noise level σ (often 10%–20% of the range between the maximum and minimum pixel values) are selected.
2. N perturbed images are then generated by adding random Gaussian noise to the original input x.
3. The vanilla gradient (sensitivity map) for each of the n perturbed images is calculated.
4. The average of these n gradients is then calculated to produce the final sharpened saliency map.

FAILURE MODES

A major failure mode that was discovered by Adebayo et. al was that some saliency methods produce visualisations that are independent of both the model's weights and the data generation process. This problem is known as the **Edge Detector Problem**. This is a problem where certain methods acted like edge detectors (a technique that requires no training data or model) rather than reflecting on what the neural network has actually learned. Since these maps often look clean and sharp, humans may mistakenly believe that they are seeing an explanation of the model's logic. However, if a method produces the same map even after a model's weights have been randomised, it is inadequate for finding any potential outliers.

Kindermans et al. argued that **many saliency methods** are unreliable because they **lack input invariance**. Input invariance is a requirement that an attribution method needs to replicate the sensitivity of the model regardless of how the input is transformed. A simple pre-processing step, such as adding a constant shift to the input data, can cause many methods to provide contrasting and incorrect attributions. Even though this shift has no effect on the model's final prediction, the saliency map changes, meaning that the explanation is sensitive to factors that do not contribute to the logic behind the model

According to what was discovered by Sundararajan et al., many existing methods fail to satisfy sensitivity and implementation invariance.

- ***Sensitivity Failure***: Many methods fail to highlight a feature that, if changed can impact the prediction score. This happens since features can flatten the activation function, meaning that they have a strong effect but a near-zero local derivative.
- ***Implementation Invariance***: This suggests that two networks which function in the same way (meaning they provide identical outputs for all inputs) should receive the same attributions. Many methods fail to do this, attributing differently based on how the network was built rather than how it behaves.

Smilkov et al. points out that **raw gradient-based maps** are visually noisy, highlighting pixels that can appear randomly to the human eye. The gradient of a neural network's output score can fluctuate sharply at very small scales. This means that the importance that is shown in a map might just be an artifact of local, meaning that variations in the partial derivatives rather than a reflection of a high-level feature like eyes or wheels

General surveys of the field note that saliency methods are often designed based on researcher intuition rather than the specific needs of the people using them

- ***Approximate Nature***: Because post-hoc methods are often approximations, they can suffer from low fidelity, failing to precisely reflect the actual operation of the original model.
- ***Surprising Artifacts***: The process of explaining saliency methods can accidentally trigger "surprising artifacts"—generating nonsensical patterns or focusing on adversarial examples that do not represent the model's behavior under normal conditions.
- ***Information Overload***: For lay-users, high-dimensional feature importance vectors can be overwhelming and redundant, describing the full decision logic rather than providing a concise, human-friendly summary.

NUMERICAL EXAMPLES

Example 1: Basic Input Gradients

The model in this example was expressed and defined using the linear equation:

$$\mathbf{y = 2x_1 + 5x_2}$$

To determine the impact of each input, the partial derivatives were then calculated. For x_1, the gradient was 2 whilst for x_2 the gradient was 5. This was defined as follows:

Using these values, we can construct the saliency vector can be constructed for the linear model as S = [2, 5]. This highlights the importance of each feature towards the model's output.

From this, we can say that x_2 has more importance than x_1. This is because when changing the value of x_2, the output that is given is affected.

This shows that **saliency is equal to the sensitivity and not the value**.

Example 2: With ReLU (for Guided Backprop)

Starting with the ReLU function **$y = \mathrm{ReLU}(2x_1 – 3x_2)$**, we'll plug in 2 and 1 for x_1 and x_2.

A **Forward Pass** was performed by placing the values of x_1 and x_2 into the linear equation $2x_1 – 3x_2$. This result was used in the ReLU model.

Since the result of the linear equation was positive, the ReLU function simply passes this value through as the final output without any clipping.

The **partial derivatives** were then calculated. For x_1, the gradient was 2 whilst for x_2 the gradient was -3.

The **Vanilla Saliency** is then defined as S = [2, 3]. This shows both the positive and negative effects of the saliency map. When computing the guided backdrop, all negative gradients are removed and thus S_{guided} = [2, 0].

Example 3: SmoothGrad

In this example, we can demonstrate **how SmoothGrad can be used to "sharpen" any noisy gradient-based maps**.

Assume the model is defined as $\mathbf{y = x_1+2x_2}$ and the input of x is [1, 1]The Vanilla Gradient is defined as S = [1, 2] and noise samples are added to the input of x.

Two noise trials are then taken:

First Trial: [+0.1,−0.1]

Second Trial: [−0.2,+0.1]

When adding the noise samples to the input, it can be deduced that the input x is close to the vanilla gradient. This shows that despite adding noise, the stable features still remain consistent, and the noise cancels itself out. After performing SmoothGrad, the final average was [1, 2].

REAL-WORLD APPLICATIONS

Saliency Maps and Input Gradient methods have a wide range of real-world applications.

One of the most important applications for these methods is to **ensure that a model is making decisions based on true evidence** rather than accidental biases in the training data. Attribution maps have revealed that some question and answering models ignore the question and rely on irrelevant words to make their predictions. To ensure that AI systems meet ethical and legal requirements, these approaches can be used to help determine whether a model is not making decisions based on biased data regarding a person's gender or background.

Saliency Maps and Input gradient methods can also be **used in safety-critical systems** where errors can have severe consequences. In self-driving cars, saliency maps can be used to help developers understand why a self-driving car fails to perform crucial actions such as braking for a stopped fire truck. For end-users, being able to see an explanation for a model's behaviour increases trust and encourages them to adopt the technology.

Saliency methods can be used **to find new discoveries and extract rules from complex models that were previously hidden in the data**. In medical case studies, models can find and interpret surprising but true patterns. For example, a model correctly identified that having asthma was associated with a lower risk of death from pneumonia—not because of the disease itself, but because those patients received more aggressive treatment. Saliency methods have also been successful applied to chemistry models and various text-based models to better understand how they process complex relational data.

Developers use attribution **to find deficiencies in a network's logic and improve its architecture**. Researchers have used saliency methods and input gradient techniques to identify specific neurons that failed to learn meaningful features, leading to improvements in network design. **Attribution facilitates the analysis of adversarial examples**, helping humans understand why a model might give an erroneous prediction with high confidence when faced with a deliberately crafted input. Methods that are truly sensitive to the model and data can be used to identify outliers in a dataset that the model is struggling to process correctly

Saliency-like mechanisms can be **integrated into various Advanced Computer Vision and NLP Tasks** such as image captioning and neural machine translation.

- ***Image Captioning***: By visually seeing the "attention weights" (a form of local interpretability), researchers can discover the parts of an image a model is "attending to" as it generates each word in a caption.
- ***Neural Machine Translation (NMT)***: These methods allow users to see how specific words in a target language depend on words in the source language during the translation process.

QUICK REFERENCE

- **Equations**

S = | ∂y/∂x|

$\widehat{M}_c(x) = 1/n \sum_{i=1}^{n} M_c(x + N(0, \sigma^2))$

- **Methods table**

METHOD	CORE MODIFICATION	PRIMARY GOAL
Vanilla	Raw Gradient	Measures local sensitivity
Guided Backprop	Discards negative gradients	Highlights features that positively activate neurons
SmoothGrad	Averaging noisy gradients	Reduces visual noise and "meaningless" local fluctuations

- **Parameters**
  - Noise σ
  - Number of Samples

- **Misconceptions**
  - Saliency is not the same as expectation.

KEY PAPERS

1. ***D. Smilkov, N. Thorat, B. Kim, F. Viégas, and M. Wattenberg, "SmoothGrad: removing noise by adding noise," arXiv:1706.03825 [cs, stat], Jun. 2017,***

In this paper, Smilikov et al. highlights the visual noise that exists in gradient-based sensitivity maps. They discovered that this noise emerges from sharp local fluctuations in the gradient of a neural network's output score, particularly in networks using activation functions like ReLU. This paper explains the SmoothGrad technique which is the average of the gradients of multiple versions of an input image perturbed with Gaussian noise. This average would produce sharper and more visually appealing heatmaps that better align with human intuition and improve discrimination across various attribution methods.

Relevance: This paper introduces the SmoothGrad technique which was one of the main methods that was explored in this project. It addresses the limitation of noise which exists in vanilla saliency maps by averaging gradients over noisy inputs. The concept is central to both the technical explanation and the implementation component of the project where SmoothGrad is used to produce clearer and more stable visualizations. This paper also supports the discussion of improving interpretability methods.

2. ***M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic Attribution for Deep Networks," proceedings.mlr.press, Jul. 17, 2017.***

In this paper, Sundararajan et al. discuss two important axioms, Sensitivity and Implementation Invariance, that effective attribution methods must satisfy. Since most of the techniques that exist fail to meet these criteria, the authors propose a method called Integrated Gradients which is designed to adhere to these axioms. Their approach requires no modification to the original network and is implemented via standard gradient calls. Through different real-world applications such as those found in image, text, and chemistry models, Sundararajan et al. showcase the method's capability of debugging networks and extracting rules. In this way, the way in which users engage with and trust deep learning models is improved.

Relevance: This paper provides a theoretical basis for attribution methods by introducing key principles such as sensitivity and implementation invariance. It is relevant to the project since it shows the limitations of gradient-based methods like vanilla saliency and guided backpropagation. This paper is used to bring up a critical discussion of what constitutes a “good” explanation, supporting the evaluation of the methods covered in this project.

3. ***J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, “Sanity Checks for Saliency Maps,” arxiv.org, Oct. 2018, Available: https://arxiv.org/abs/1810.03292***

In this paper, Adebayo et al. propose a strict methodology to discuss the reliability of saliency methods, warning that reliance on visual appeal alone can be misleading. Their primary contribution is the introduction of saliency checks – specifically randomising model weights and data labels – to determine if explanations are tied to the model’s logic. They demonstrate that several popular methods produce visualizations that are independent of the model and the data, functioning essentially as simple edge detectors. These methods are useful in critical tasks like debugging models or identifying outliers.

Relevance: This paper is crucial for the failure modes section of the project. It demonstrates that saliency maps can remain visually similar even when the parameters of the model are randomized. This suggests that the explanations may not reflect the behaviour of the learned model. This finding directly supports the critical analysis of saliency methods and reinforces the idea that visual explanations must be interpreted with caution.

4. ***P.-J. Kindermans et al., “The (Un)reliability of saliency methods,” arXiv.org, 2017. https://arxiv.org/abs/1711.00867***

In this paper, Kindermans et al. analyses the technical weaknesses that exist within current attribution techniques, arguing that they are often sensitive to factors that do not influence model predictions. This paper introduces the input variance rule, which states that a saliency method must replicate the model's sensitivity even when the input undergoes changes that do not affect the output. This paper proves that when constantly shifting the input data, many saliency methods can produce incorrect results.

Relevance: This paper highlights how unstable saliency methods can be by demonstrating that simple input transformations can significantly affect saliency maps without affecting the predictions of the model. This paper talks about multiple failure modes particularly regarding sensitivity and lack of robustness which are relevant to the project. The paper strengthens the argument that gradient-based explanations may not always be reliable indicators of feature importance.

5. ***M. Du, N. Liu, and X. Hu, "Techniques for Interpretable Machine Learning," arXiv:1808.00033 [cs, stat], May 2019, Available: https://arxiv.org/abs/1808.00033***

In this paper, Du et al. provides a in-detail survey and a systematic framework for understanding how to explain complex "black-box" models. Their primary contribution is a taxonomy that categorizes interpretability into intrinsic versus post-hoc methods, and global versus local scopes. The paper specifically describes the need to have faithful evaluation metrics, and it also describes a proposed transition toward user-centric, human-friendly explanations. By using different types of explanations, the authors offer guidelines to make AI systems more reliable and trustworthy for end-users.

Relevance: This paper provides a good explanation of interpretability techniques, placing saliency methods within the wider context of explainable AI. It provides reasons to explain why interpretability is important which is relevant to the introduction and background sections of the study notes. It also supports comparisons and discussions about when saliency methods are appropriate to use.

6. ***L. Itti, C. Koch, and E. Niebur, "A model of saliency-based visual attention for rapid scene analysis," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 20, no. 11, pp. 1254–1259, 1998, doi: https://doi.org/10.1109/34.730558.***

In this paper, Itti et al. introduces a biologically inspired computational framework for bottom-up visual attention. The main contribution of this paper is a system that decomposes input into topographical feature maps—colour, intensity, and orientation—using multi-scale centre-surround operations to detect local discontinuities. These features are integrated into a single master saliency map through a unique normalization process that promotes conspicuous locations while suppressing noise. The model would sequentially select attended regions in order of decreasing saliency. This efficient approach successfully replicates human search behaviour across synthetic and complex natural imagery.

Relevance: This paper provides the conceptual origin of saliency and is relevant to the biological foundations section of the study notes.

7. ***L. Itti and C. Koch, "A saliency-based search mechanism for overt and covert shifts of visual attention," Vision Research, vol. 40, no. 10–12, pp. 1489–1506, Jun. 2000, doi: https://doi.org/10.1016/s0042-6989(99)00163-7.***

In this paper, Itti and Koch provide a detailed computational implementation for bottom-up attentional selection. The main contribution of this paper is a within-feature spatial competition scheme that is inspired by non-classical neural interactions. This paper solves the challenge of integrating 42 disparate feature maps (color, intensity, orientation) into a single saliency map. By iteratively applying Difference-of-Gaussians (DoG) filtering, the system improves the isolated conspicuous peaks while suppressing uniform background noise. The model successfully replicates human psychophysical search behavior and outperforms human observers in detecting camouflaged targets in high-resolution natural scenes.

Relevance: This paper links saliency with attention mechanisms and perception and is relevant to the biological foundations section of the study notes.